

Application Guidebook

Engine Management System (ECU 9) Diesel Engine
for C&I & O&G applications

Emission-related Installation Instructions

10V1600Cx0

12V1600Cx0

MS65037/02E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
10V1600C50	57 kW/cyl.	5B, continuous operation, variable
10V1600C60	57 kW/cyl.	5B, continuous operation, variable
10V1600C70	61 kW/cyl.	5B, continuous operation, variable
12V1600C50	53 kW/cyl.	5B, continuous operation, variable
12V1600C60	57 kW/cyl.	5B, continuous operation, variable
12V1600C70	61 kW/cyl.	5B, continuous operation, variable

Table 1: Applicability

Preface

This document is exclusively intended for use by

- Integrators
- Vehicle manufacturers
- Regional sales centers and distributors.

It includes general guidelines and information on the interfaces of the Engine Control Unit ECU 9.

Ensure that the most recent version of the Application Guidebook is available.

Please note that this manual is not exhaustive in all areas and reflects the current state of development. The content of this Application Guidebook shall not be used as a basis for any form of legal claim.

They in no way relieve the manufacturer of the vehicle / fully integrated system of his obligation to plan, execute and check correctly.

Values are shown by way of example for purposes of illustration only. Refer to the documents included in the scope of delivery for details of components, applicable values and dimensions.

Follow the instructions provided in the Safety section (→ Page 9) of this document, and in the Operating Instructions of the engine concerned (see drawing list and document list section (→ Page 96)) and its component parts, to avoid hazardous operating states.

Table of Contents

1 General Information		7 Electrical Connection to Vehicle/Unit Interface	
1.1 General information	7	7.1 General information	52
1.2 Installation Description - Use	8	7.2 ECU 9 - Maximum scope of functions	53
2 Safety		7.3 Sales Code overview for ECU 9 configuration	55
2.1 Important provisions for all products	9	8 Emergency Stop Safety System	
2.2 Important provisions for Series 1600 engines in C&I and O&G applications	11	8.1 General information	60
2.3 Intended use of Series 1600 engines in C&I and O&G applications	13	8.2 Emergency stop by IGI (TL15) input	61
2.4 Personnel and organizational requirements	15	8.3 Emergency stop by ESI input	62
2.5 Initial start-up and operation - Safety regulations	16	9 CAN Control	
2.6 Assembly, maintenance, and repair work - Safety regulations	18	9.1 CAN control	63
2.7 Fire and environmental protection, fluids and lubricants	22	9.2 CAN J1939 nodes for Engine Control Unit - Setting	64
2.8 Standards for warning notices in the text and highlighted information	24	10 Cabling	
3 Engine Management System - General Information		10.1 Wiring and connection - General information	65
3.1 Electrical system - General information	25	10.2 Plant cable W001 - ECU 9 at connector X1	66
3.2 Engine management system - Use	26	10.3 ECU 9 - Connector X1	70
3.3 Information for the vehicle manufacturer	28	11 Sensors, Actuators and Injectors on Engine Wiring Harness	
3.4 Monitoring and protection by the engine management system	29	11.1 Sensors and actuators - Overview	73
4 Engine Control Unit ECU 9		11.2 ECU 9 connector X2	75
4.1 Function	31	11.3 ECU 9 connector X3	77
4.2 ECU 9 - Mechanical design	32	12 Starter Cabling	
4.3 Engine Control Unit ECU 9 - Technical data	34	12.1 Engine start via ECU 9	78
4.4 ECU 9 - Operating voltage	36	12.2 Engine start via external control	79
5 Binary Inputs		13 Electronic Engine Identification Label EIL	
5.1 Binary inputs - General information	41	13.1 General information - EIL	80
5.2 Binary inputs - Software setting	43	13.2 EIL - Configuration	81
6 Grounding Concept ECU 9		13.3 EIL - Function	82
6.1 General information	44	13.4 EIL configuration in case of replacement	83
6.2 Grounding concept for ECU 9 inputs and outputs	46	14 Interfaces	
		14.1 Diagnostic interface	84

15 Diagnostic System DiaSys®		16.2 Insulation testing (ground testing)	95
15.1 Diagnostic system DiaSys® - General information	85	17 Appendix	
15.2 DiaSys® - Scope of delivery	86	17.1 Drawing list and document list	96
15.3 Notebook/PC	87	18 Appendix A	
15.4 Connection of DiaSys® to ECU 9	88	18.1 List of abbreviations	97
15.5 Software level for ECU 9	89	18.2 MTU Contact/Service Partners	98
15.6 Settings - DiaSys®	90	19 Appendix B	
16 EMC and Grounding		19.1 Index	99
16.1 EMC regulations	94		

1 General Information

1.1 General information

A large number of measured variables and control signals have to be acquired, evaluated and processed in order to control and regulate a diesel engine. This information, which determines how the diesel engine responds, may originate either from the engine (e.g. operating data such as pressures and temperatures) or from the so-called plant, i.e. from the vehicle or unit.

Engine Control Unit ECU 9 interfaces between the plant and the sensors/actuators on the engine.

Plant inputs are:

- Engine start
- Engine stop
- Engine speed or engine torque

Actuators on the engine are:

- Injectors for the individual cylinders

Moreover, the Engine Control Unit also has to acquire and evaluate additional engine operating data and ambient data, as these factors influence the way in which the engine responds to meet the requirements of the plant. Additional sensors and actuators are installed on the engine for this (e.g. pressure and temperature sensors, and exhaust gas sensors, flaps and actuators). The Engine Control Unit controls all these actuators as well in order to meet the demands made on the engine other than speed and torque. These are e.g. emission values (density, particle and NOx emissions), fuel consumption, smooth running etc.

1.2 Installation Description – Use

This installation description is intended for companies (OEMs) dealing with the integration of MTU Series 1600 engines in vehicles or units. Following these guidelines carefully is an essential contribution to optimizing operation of the vehicle/unit and its component parts both from the technical and economical viewpoints. The engine manufacturer shall not be held liable for any defects or inadequacies in the fully integrated system resulting from failure to heed this guideline

Applicability

This Installation Description applies exclusively to Engine Control Unit ECU 9 featuring software version 2.0 or 2.1 and parameter configuration level 1b.

Only use parts/components which have been approved by the engine manufacturer. Consult the engineering department before using any other parts/components.

The OEM shall ensure proper functioning of the vehicle/unit, e.g. stability of components in the fully integrated system (vehicle/machine).

The OEM is responsible for providing objective evidence of functionality and lawful operation.

Publication

This Installation Description is published online.

The Installation Description is regularly updated to reflect ongoing development. Thus only the version published online is to be considered valid. The user is responsible for ensuring that only the latest version is consulted.

Printed versions are not subject to change management.

Trademarks

Use of MTU trademarks on products manufactured by the OEM requires prior approval from MTU.

®	® indicates a registered trademark.
©	© indicates a copyrighted name.
TM	® indicates an unregistered trademark.

Table 2: Trademark symbols

2 Safety

2.1 Important provisions for all products

General information

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number and must match with the information on the title page of this manual.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. When operating the machine in the EU: The second nameplate must be affixed in a prominent position as described in the accompanying specifications.

Emission regulations and emission labels

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 13).

Emission certification applicable to engines with EPA Nonroad Tier 4 emission certification in accordance with 40 CFR 1039

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph described the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emission labels

On all MTU engines fitted with emission labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emission labels:

- The relevant emission labels must be affixed to the spare part.
- Emission labels shall not be transferred from the replaced part to the spare part.
- The emission labels must be removed from the replaced part and destroyed.